A mathematical explanation of empirical results in machine learning

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Context

Modern machine learning achieves outstanding predictive performance, but classical statistical learning theory [6, 10] fails to explain many of its successes [2, 4].

Example

Generalization ability of over-parametrized models vs. classical overfitting arguments.

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$\star$ We need new mathematical tools to explain successful results $\star$

Why should we bother?

- **scientific curiosity**
- **explainability**
- **future developments**

"*nothing is more practical than a good theory*" [9]

State of the art

Current partial progress, on three fronts [4, 8]:

- **approximation theory**: why deep nets represent complex functions well [5]
- **optimization theory**: why non-convex objectives are minimized effectively [1]
- **generalization theory**: why interpolating models still perform well [7]

$\Rightarrow$ no unified theory yet matching the breadth of empirical practice.

Research objective

Research question

Fill the **gap between practice and theory** in deep learning

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How I intend to do it:

1. **mathematical framework** and hypotheses
2. test hypotheses with **experiments**
3. if discrepancies: **develop the theory** further, then retest
4. if not: run **more complex experiments** to test robustness

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- *definition* $\rightarrow$ *theorem* $\rightarrow$ *proof* [11, 4]
- mental pictures (just enough theory)
- alternation between theory and practice

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Research question

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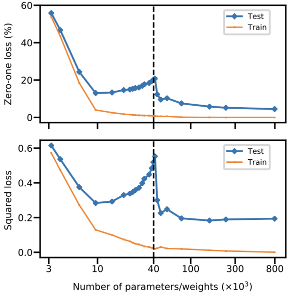


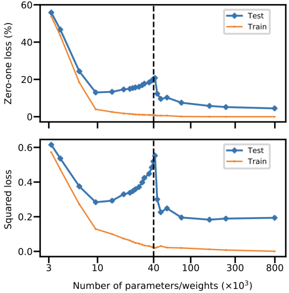
Double descent - Belkin et al. [3]

$\rightarrow$ Analysis discrete update vs. continuous-time flow (implicit regularization)

Example: the puzzle of generalization

Research question

In the context of gradient-based learning algorithms, **why do over-parametrized models generalize better**?



Double descent - Belkin et al. [3]

→ Analysis discrete update vs. continuous-time flow (implicit regularization)

- mathematical formalization of the problem
- experiments on gradient-based algorithms dynamics
- formulate and test hypotheses
- apply framework to test implicit regularization argument

Thank you

Questions?

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Appendix I

Proposition [Comparing discrete and continuous trajectories]

For a step size $\alpha \in \mathbb{R}_{>0}$, $t \in \mathbb{R}_{>0}$

$$\theta_{t+\alpha} = \theta_t - \alpha \nabla \ell(\theta_t) \quad \text{and} \quad \frac{\mathrm{d}\theta}{\mathrm{d}t} = -\nabla \ell(\theta(t))$$

Up to order α, gradient descent corresponds to gradient flow of:

$$\ell_{discr=contin}(\theta) = \ell(\theta) + \frac{\alpha}{4} \left\| \nabla \ell(\theta) \right\|^2$$

Proof:
Consider the usual (discrete-time) update rule:

$$\theta_{\tau+1} = \theta_\tau - \alpha \nabla \ell(\theta_\tau)$$

for $\tau \in \mathbb{N}$, $\alpha \in \mathbb{R}_{>0}$ step size.

Appendix II

Let $t = \alpha\tau$ and consider the finite-difference

$$\theta_{t+\alpha} - \theta_t = -\alpha \nabla \ell(\theta_t).$$

Now, taking the limit we obtain the continuous-time update rule (where we omit the dependence of θ on t to ease notation)

$$\lim_{\alpha \to 0} \frac{\theta_{t+\alpha} - \theta_t}{\alpha} = \frac{\mathrm{d}\theta}{\mathrm{d}t}.$$

Then,

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\nabla \ell(\theta). \tag{1}$$

Allowing

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\nabla \ell(\theta) + \alpha g(\theta),$$

we want to find the correction term g such that the continuous and discrete trajectories agree up to order α. Let us consider the

Appendix III

function $\theta(t)$ and inspect its Taylor expansion starting from $\theta_0 = \theta(0)$, for a jump $\Delta t = \alpha$ (1 iteration in discrete sense):

$$\begin{aligned}
\theta(t) &\approx \theta_0 + \alpha \frac{\mathrm{d}\theta}{\mathrm{d}t} + \frac{1}{2}\alpha^2 \frac{\mathrm{d}^2\theta}{\mathrm{d}^2 t} \\
&= \theta_0 - \alpha \nabla \ell(\theta) + \alpha^2 g(\theta) + \frac{1}{2}\alpha^2 \frac{\mathrm{d}}{\mathrm{d}t}\Big(- \nabla \ell(\theta) + \alpha g(\theta) \Big) \\
&\approx \theta_0 - \alpha \nabla \ell(\theta) + \alpha^2 \left(g(\theta) + \frac{1}{2}\left(- \nabla^2 \ell(\theta) \frac{\mathrm{d}\theta}{\mathrm{d}t} \right) \right) \\
&= \theta_0 - \alpha \nabla \ell(\theta) + \alpha^2 \left(g(\theta) + \frac{1}{2}\left(\nabla^2 \ell(\theta) \Big(\nabla \ell(\theta) + \alpha g(\theta) \Big) \right) \right)
\end{aligned}$$

where the terms in pink are of 3rd order in α, hence disregarded. Then, we impose $\theta(t) = \theta_{\tau+1}$, obtaining

$$g(\theta) = -\frac{1}{2}\nabla^2 \ell(\theta) \nabla \ell(\theta).$$

Appendix IV

Hence, the following continuous-time update rule agrees with the discrete-time update up to order α

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\nabla \ell(\theta) - \frac{1}{2}\alpha \nabla^2 \ell(\theta) \nabla \ell(\theta). \tag{2}$$

Lastly, we notice that this is the same as applying the usual continuous-time update rule to the modified loss

$$\tilde{\ell}(\theta) = \ell(\theta) + \frac{1}{4}\alpha \nabla \ell(\theta)^t \nabla \ell(\theta) = \ell(\theta) + \frac{1}{4}\alpha \left\| \nabla \ell(\theta) \right\|^2 \tag{3}$$